SYLABUS

Precalculus :

Hi i hope you good and enjoy what i'm write this subject is a good breach between algebra , geometry and calculus

Chapter 1

1.1 Basic Idea of Function and Transformation Theorem

Function

Fundamental tools in a lot of framework of mathematics represent rule that change every input into another output by apply basic mathematical rule such us (adding, multipication, subtraction, divided and etc). Let starting with humble equation in math:

$$f(x) = x, which\ x\ act\ as\ domain\ and\ f\ as\ rule\ that\ operation\ for\ evey\ input\ allow$$

Core concept of function is every unique input give excatly one unique output. Which means difference input give same output allow but one input can't have more than one output.

Vertical Test

Best way to test function by vertical test, graphically if vertical you made across more then one point it false as a function.

Transformation Theorem

Transformation represent as function. Notice that:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \rightarrow A.\boldsymbol{x} = \boldsymbol{x}'$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = b_1 \rightarrow R_1.\boldsymbol{x} = b_1$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = b_2 \rightarrow R_2.\boldsymbol{x} = b1$$

Every row of matrix A represent distinct function which change vector x into vector b as (**x'**). In linear algebra can represent as row picture or column picture, $\begin{bmatrix} x \\ y \end{bmatrix}$ and $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ represent point in xy plane of R^n space. Instead, $\begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ represent mathematical rule for proces every input by scalar multiplication form an equation.

In easy way, $(x, y) \rightarrow_{y=x} (y, x)$ proof that transformation is function because statisfies for domain, range and rule of function.

A good idea of transformation is how to see function or imagine it by graphically as the behavior of transformation instead numerical function that count to predict another point.

Rigid Transformation

Function that preserve geometrical shape such us (translation, reflection and rotation)

Translation

Y-axis: $f(x)' = f(x) \pm c$

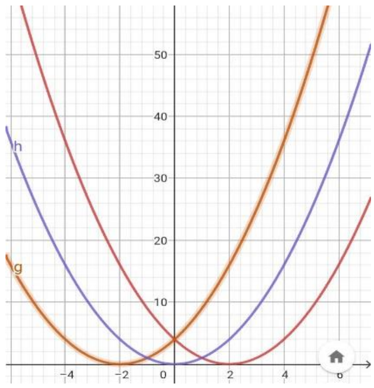
X-axis: $f(x)' = f(x - c)$

If c>0 function move to the left. Hence, if c<0

The function would move to the right

Example: $x^2 \rightarrow (x-2)^2$

Then, $x^2 \rightarrow \left(x-(-2)\right)^2$



Notice that for first function we need subtitute opposite value to get zero as the output. Such us: $(x-2)^2 = (2-2)^2 = 0$

This concept difference if we're talk about translation along y-axis, because basicly f(x) represent general answer for this behavior. Remember what f represent ? It's the output that dependt on x as an input, meaning to get y intersecpt (x=0) we fon't need modification the y value and because x Independent variable we can just plug zero in the equation. This difference with x-axis, to get x intersecpt (y=0) we need manipulation x value to get zero as the result, this manipulation make the graph move invertibley.

Reflection

In general, a lot of function can represent as reflection function.

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix} \rightarrow (y - axis)$$

Such us $(x^2,\ \cos(x)\,, |x|, \text{and etc})$

You can graphically imagine how the function looks like, notice that the distance of preimage and image to y-axis a s vertical mirror line always same.

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix} \rightarrow (x - axis)$$

Circle Equation is one of common example, $(x \pm a)^2 \pm (y \pm b)^2 = r^2$

In general, a lot of function on this behavior fail in vertical test but, if we're use separately with piece wise function and represent other piece as linear transformation it be a function.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix} \rightarrow (y = x)$$

A good example in this transformation is an inverse function, this transformation explain the relationship of exponential ($f(x)$)and Logarithmic ($f(x)^{-1}$) graphically

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix} \rightarrow (y = -x)$$

Any exponential function which power an negative odd number use this transformation to explain the behavior of function:

$$f(x) = x^n, which\ n = -(2x + 1)$$

Another transformation:

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix} \rightarrow (y = 0)$$

This transformation can represent derivative function of quadratic function, let:

$$f(x) = x^2$$

$$\frac{d}{dx} f(x) = 2x$$

Rotation

Intuitively, we can see linear transformation of reflection as rotation viewpoint. Let a point floating at xy plane the distance to origin point give vector at point A. Rotation at certain angle give another point which distance to specific line equals both for preimage and image.

$$A(x, y) = \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \ A' \ (x', y') \text{ by}$$

$$[\cos(a) \quad \sin(a)] \times \begin{bmatrix} x \\ y \end{bmatrix} = x'$$

$$[-\sin(a) \quad \cos(a)] \times \begin{bmatrix} x \\ y \end{bmatrix} = y'$$

Non Rigid Transformation

This transformation change the shape of any function by multiplication of scalar.

Dilatation

The function would stretch or compress vertically and horizontally. Basicly, stretch is the condition while function get increase fastly far from opposite coordinat axis and get closer to another coordinat axis depend on vertically or horizontally. Such us vertical dilatation would and if increase then it move fastly from x Axis and get closer to vertical (y) axis.

$$f(x)' = \pm c \times f(x)$$

If c>1 then the function would stretch and increase/decrease fastly in vertical axis

If 0<c<1 then the function would compress ad increase/decrease slowly in vertical axis

Horizontally

$$f(x)' = f(\pm c.x)$$

If c>1 then the function would compress because function increase/decrease slowly in horizontal axis

If 0<c<1 then the function would stretch because the of function increase/decrease fastly in horizontal axis

Notice that there's pattern in basic of transformation theorem,did you notice it ?

Outside vs Inside Prathecies

Represent two distinct way to see function graphically and it's transformation.

Vertical viewpoint

Denoted by,

$$f(x)' = A * f(x)$$

The symbol of * represent basic arithmetic operation (+, -, c, 1/c), the behavior of function and Transformation of function statisfies for general transformation such us:

$$if\ (k + f(x))\ represent\ up$$

$$if\ (k - f(x)) represent\ down$$

$$if\ (c \times f(x)) increase$$

$$if\ \left(\frac{1}{c} \times f(x)\right) decease$$

Horizontal viewpoint

We threat horizontal axis as output (range) value instead input (domain) and vertical axis as domain axis to determine how fast function increase or decrease dependent on input (y-axis). Remember some expression increase/decrease fastly if it stretch and get closer to main axis in this case is (x-axis as horizontal line). This make basic arithmetic operation behave oppositely such us:

$$if\ \left(f(x+c)\right)\ move\ to\ negative\ value$$

$$if\ (f(x-c) move\ to\ positive\ value$$

$$if\ \left(f(c.x)\right)\ compress\ and\ increase\ or\ drcrease\ slowly$$

$$if\ \left(f\left(\frac{1}{c}.x\right)\right) stretch\ ad\ incerase\ or\ decrease\ fastly$$

1.2 Basic ideaof calculus.

Key concept

Calculus is mathematical branch that study of continuous change. If in basic Algebra we're study on discrete change in calculus we threat function as point that move and change continuously. For example, function of position give us information what position are we at certain time but not close enough to explain how fast we change position at that certain time. Think about car and the speedometer and let said one of your family measurement your position every single time. Your family plot graph that tell you how your position any single time but don't know your velocto move at that certain time basicly they don't know the value that show at your speedometer. Calculus, specially derivative stuff tell you how your Y value change at fix point of X as input of function by apply concept limit movement.

So basicly derivative provide a good understanding to make a new graph that tell you how curve yout original function are. This explain famous word "rate of change"

Example,

$$s(t) = 2t^3 \qquad (1)$$

$$\frac{d}{dt}s(t) = v(t) = 6t^2 \qquad (2)$$

$$\frac{d}{dt}v(t) = a(t) = 12t \qquad (3)$$

Second function explain how my position change of time and third function represent how my velocity change dependent of time. Second function represent as the function to determine rate of change or slope of cubic function at any time. At the end, third function represent how curve second function looks like.

1.3 Linear Equation

Core concept

Linear equation is constant rate of change meaning every output coresponding of the point at line of input axis (horizontal line).

General form

$$\text{Basic equation: } y = mx + c$$

$$\text{Standard form: } Ax + By = C$$

$$\text{Point} - \text{slope equation: } y - y_1 = m(x - x_1)$$

Linear equation represent a straight line, the graph of rate of change would be a constant line for every input or value. Commonly, use in specific world problem that involved no changes such us battery capacity for one week the reason why i limit the interval of t is because capacity of battery would starting decrease as t increase. Interesting stuff here is we can use standard form to solve system of equation

Example,

$$2x + 3y = 5$$

$$5x + 7y = 8$$

Change into basic equation term,

$$y_1 = -\frac{2}{3}x + \frac{5}{3}$$

$$y_2 = -\frac{5}{7}x + \frac{8}{7}$$

Remember in solving linear equation we search point (x,y) that statisfies for map (x,y) into b vector in form of,

$$\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

Meaning (x,y) is same value for both function. Then,

$$y_1 = y_2$$

$$-\frac{2}{3}x + \frac{5}{3} = -\frac{5}{7}x + \frac{8}{7}$$

$$\left(-\frac{2}{3} + \frac{5}{7}\right)x = \frac{8}{7} - \frac{5}{3}$$

$$\frac{1}{21}x = -\frac{11}{21}$$

$$-\frac{2}{3}x+\frac{5}{3}=-\frac{5}{7}x+\frac{8}{7}$$

$$-\frac{2}{3}\times -11+\frac{5}{3}=-\frac{5}{7}\times -11+\frac{8}{7}$$

$$3^2=3^2$$

Then solution of system of equation is (-11,9) which statisfies for,

$$\begin{bmatrix}2 & 3\\5 & 7\end{bmatrix}\times\begin{bmatrix}-11\\9\end{bmatrix}=\begin{bmatrix}5\\8\end{bmatrix}$$

$$\begin{bmatrix}R_1\\R_2\end{bmatrix}.\boldsymbol{x}=\boldsymbol{b}$$

1.4 Plynomial

Plynomial is mathematical term with variable, cofficient and constant.

General form: $P(x)=\ a_1x^n+a_2x^{n-1}+\cdots+a_nx^{n-n}$

Basicly plynomial is a function with power is positive integer greater than equals zero. Plynomial can't have negative or fraction on the power and the domain is statisfies for all real number. The reason why plynomial can't have fraction or negative value is because the limit and domain problem. The reason why it have to define in all real number is because plynomial statisfies for closed operation which (addition, subtraction, and multiplication) divided is not involved because in advance math there's no divided operation, it define as multiplication on denominator form such us $(\frac{1}{2},\frac{1}{4},\frac{1}{8},\dots,\frac{1}{n})$ so, for every mathematical rule that statisfies closed operation Mus be statisfies for:

$$x^n\ defined,$$

$$ax^n\ defined,$$

$$ax^n\pm by^c\ defined$$

Defined means has an output for every real number that plug in that three specific rules. So mathematician avoid negative or fraction for power to avoid undefined in three rules.

Power

Basicly, because power for any plynomial an integer then we can defined it to three distinct kind: it's zero, even and odd. Difference kind of power give difference behavior of function.

$$P(x)=x^{n-n}$$

Represent a straight line across point (0,1) why $0^0=1$?

Remember, universally it's equals 1 but in limit or analycis real it's represent undefined, Basicly there's no rigorious proof of this problem but combinatory solve it intuitively. Let,

$$a^b = 0^0 = 1$$

a is the set of variable and b is n-tuple way such us a is your sock and b is the way you organized and match your sock and the only way to organized zero think to zero way is one way which is zero element of set. If you study about set theory

Let, $A = \{a\}$ your variable in that set wasn't a which is one but 2^n , n represent how many variable are there in that set which 2 (a, {}) or zero or set of nothing. That "nothing" is the only thing is there in a^b which you organized and match but you just have on thing that means you got only one way which is,

$$0^0 = 1$$

Represent the way to get 0-tuple of 0 is 1 way.

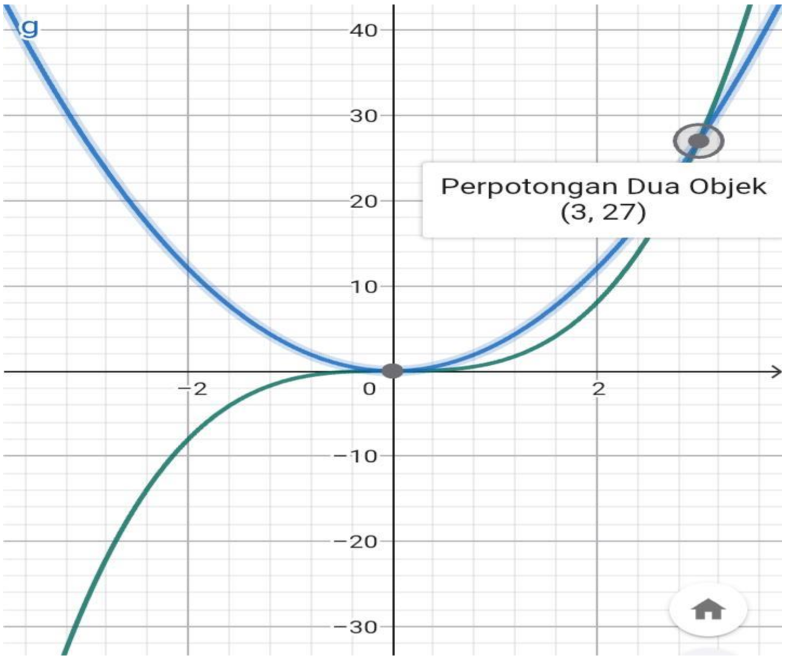
$$P(x) = x^{2n}$$

If it an even number the graph would seen as a curve which can stretch or compress by linear transformation. It's act as reflection to y axis because two opposite value give a same output. The rate of change or function of slope (derivative) would seen as part of sin (x) wave function because it's (decrease -increase)

$$P(x) = x^{2n+1}$$

If it's an odd number the graph would seen as sin (x) at interval $\left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right)$ basicly an odd function represent linear transformation in interval $\left(-\frac{\pi}{2} \leq x \leq 0\right)$ by y=0 or rotation at 180 degree conter clockwise or clockwise depend on what interval you fix is that $\left(-\frac{\pi}{2} \leq x \leq 0\right)$ or $\left(0 \leq x \leq \frac{\pi}{2}\right)$. Graphically the rate of change in odd plynomial function would be (increase -constant -increase).

For an example, green line represent x^3 and blue line represent 3x^2 notice that graphically the green line can Stretch in y axis as scalling by dilatation of transformation and the blue line represent the rate of change, let said you observe your friend with his motorcycle by the position in anytime then you got the pattern, position=y=t^3 then you can find the change of you friend position in any certain time by d/dx y=3t^2 you plot on graph because in real life there's no negative value for ride at least for know. Theb you can conclude that he increase velocity starting from t=0 and so on.



Proerties of Symetricity

Did you notice something in plynomial function ?

$$\text{Let: A} \rightarrow [f(x) = f(-x)]$$

$$\text{B} \rightarrow [-\text{f}(\text{x}) = \text{f}(-\text{x})]$$

Then,

$$\frac{d}{dx}A = B, \frac{d}{dx}B = A$$

Example;

$$f(x) = x\text{^}4$$

$$\frac{d}{dx}f(x) = 4x^3 \rightarrow \ odd\ function$$

$$g(x) = x^3$$

$$\frac{d}{dx}g(x) = 3x^2 \rightarrow \ even\ function$$

This relationship can also see graphically in their function and slope function

Why ?

IT because we do exponential derivative operation, in general all function can represent as exponential function so to make easy mathematician form derivative function in exponential function form

$$\frac{d}{dx}f(x) = nx^{n-1}$$

Patternly, even and odd number arrange in $2n$ and $(2n + 1 = \text{even} + 1)$, this give us conclution that even and odd number appear switchly such us:

$$0, 1, 2, 3, \ldots, 2n, 2n + 1$$

Then,

$$odd = even + 1$$

$$even = odd - 1$$

Both sequence and two general equation determine operation,

$$even + 1 = odd$$

$$even - 1 = odd$$

$$odd + 1 = even$$

$$odd - 1 = even$$

Then,

$$\frac{d}{dx}x^{2n} = (2n)x^{2n-1} = (2n)x^{even-1} \rightarrow \ odd\ function$$

$$\frac{d}{dx}x^{2n+1} = (2n + 1)x^{2n} = (2n + 1)x^{odd-1} \rightarrow even\ funtion$$

Another rigid proof, by using the linear transformation of even, odd function and chain rules:

Basic idea before proof, every function can derivative using equation :

$$\frac{d}{dx}f(x) = nx^{n-1}$$

Chain rule basicly act as derivative in Composition function

Example

$$\frac{d}{dx}f(x) = \frac{d}{dx}f[(x - 2)^2] = \frac{d}{dx}[u^2 o\ (x - 2)] = \frac{d}{dx}u^2 o \frac{d}{dx}(x - 2) = 2u\ o\ 1$$
$$= 2(x - 2) \times 1 = 2x - 4$$

$$\frac{d}{dx}f(x) = \frac{d}{dx}[(x^2-8)^4] = \frac{d}{dx}[u^4 o (x^2-8)] = \frac{d}{dx}u^4 o \frac{d}{dx}(x^2-8) = 4u^3 o\ (2x)$$
$$= 8x(x^2-8)^3$$

Proof 1,

$$f(x) = f(-x)$$

$$\text{Let}, f(-x) = k(x)$$

$$k(x) = k(-x) \rightarrow\ f(x) = f(-x)$$

Because k of x have excatly output of k of -x i can rewrite that:

$$k(x) = k(-x) = f(-x)$$

$$f(x) = k$$

$$\frac{d}{dx}f(x) = \frac{d}{dx}k$$

$$\frac{d}{dx}f(x) = \frac{d}{d(x)}k(-x) = \frac{d}{dx}[k(-x)\ o - x] = \frac{d}{dx}[f(-x)\ o - x] = \frac{d}{dx}f(-x)\ o\frac{d}{dx}(-x)$$

$$\frac{d}{dx}f(x) = \frac{d}{d(x)}f(-x) \times -1$$

$$\frac{d}{dx}f(x) = -1 \times \frac{d}{dx}[f(-x)]$$

$$f'(x) \times -1 = -1 \times -1 \times f'(x)$$

$$-f'(x) = f'(x)\ similiar\ as - f(x) = f(x) \rightarrow\ odd\ function$$

Proof 2,

$$-f(x) = f(-x)$$

$$\text{Since now}, \frac{\mathrm{d}}{\mathrm{dx}}\mathrm{f(x)} = \mathrm{f'(x)}$$

$$-f'(x) = f'(-x)\ o\ (-x)'$$

$$-f'(x) =\ f'(-x) \times\ -1$$

$$-1 \times -f'(x) = f'(-x) \times -1 \times -1$$

$$f'(x) = f'(-x)\ similiar\ as\ f(x) = f(-x) \rightarrow\ even\ function$$

Extend and Factorize Form

I sure you all know basic equation in algebra school

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

That equation statisfies for plynomial function. Then:

$$P(x) = a^2.x^0 \pm 2ab + a^0 b^2$$

Which $\text{degree} = \mathrm{n}^2, \text{variable} = (\mathrm{a}, \mathrm{b}), \text{coefisien} = (1, -2, 1), \text{and constant} = \mathrm{b}^2$ meaning every extended form binomial equation statisfies for standar form of plynomial. Then:

$$\text{standard form: } P(x) = (a \pm b)^n$$

$$a_1 x^n \pm a_2 x^{n-1} \pm \cdots \pm a_n^0 = \text{ Extended form of binomial}$$

$$a_1 x^n \pm a_2 x^{n-1} \pm \cdots \pm a_n^0 = a^n x^0 \pm .. On \pm x^0 b^n$$

Then extended formula statisfies to represent standard form of plynomial function. It's difference with factorized. Let use basic equation in algebra before in certain way, let:

$$P(x) = (a - b)^2 = a^2 - 2ab + b^2$$

Factorized plynomial function make function more simple, if plynomial function live in vector space then factorized form must be live in vector space proof by factorized form can rebuilt again into plynomial function. If plynomial in term of factorized then:

$$a^2 \mp 2ab = -b^2 \rightarrow a^2 \mp 2ab + b^2 = 0 \rightarrow (a \pm x)(a \pm y) = 0$$

Then etleast one of the factory in factorized form must be equals zero to producte zero output.

Example: $(x - 2)(x + 3) = 0$ let $\mathrm{x} = 2$ then

$$(2 - 2)(2 + 3) = 0(5) = 0$$

Techniques of Symplification

$$(a - b)^{2^n} = \left(a^{\frac{2^n}{2}} + b^{\frac{2^n}{2}}\right)\left(a^{\frac{2^n}{2}} - b^{\frac{2^n}{2}}\right) = \left(a^{2^{n-1}} + b^{2^{n-1}}\right)\left(a^{2^{n-1}} - b^{2^{n-1}}\right)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp 2ab + b^2)$$

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}, \text{staisfies for quadratic funtion to}$$

$$\text{get rational, irational, and imaginer solution}$$

$$(a + b)^{2^n} = \left(a^{\frac{2^n}{2}} + (bi)^{\frac{2^n}{2}}\right)\left(a^{\frac{2^n}{2}} - (bi)^{\frac{2^n}{2}}\right) = \left(a^{2^{n-1}} + (bi)^{2^{n-1}}\right)\left(a^{2^{n-1}} - b^{2^{n-1}}\right)$$

$$If\,\frac{2^n}{2} = 2^{n-1} = \ odd\ number\ which\ i^{2^1} = i^{2^3} = i^{2^{2n+1}} = -1$$

I hope you enjoy this paper as well i enjoy Mathematics much. I Will post other paper next time. I'm so sorry because i can't post set of problem and solution right now. Hope you enjoy and have a good day!

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